

# XI(SEAN) CHEN

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## EDUCATION

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| <b>University of Massachusetts Amherst</b> <ul style="list-style-type: none"><li>• PhD in Computer Science (GPA: 3.9/4)</li></ul> | <b>Amherst, MA</b><br>2019-2025     |
| <b>Central South University</b> <ul style="list-style-type: none"><li>• B.E. in Computer Science (GPA: 3.9/4)</li></ul>           | <b>Changsha, China</b><br>2014-2018 |

## EXPERIENCE

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| <b>Pretraining &amp; Scaling – Amazon AGI Foundations</b><br>Applied Scientist <ul style="list-style-type: none"><li>• Optimized pre- and mid-training recipes by scaling law for Amazon’s base models (<b>hundreds B, 10k+ gpus</b>).</li><li>• Diversified reasoning path by contrastive learning and self-supervised token-efficient curriculum for mid-training.</li><li>• Unified datamix within a framework that stabilizes adaptive attention and distribution-constrained optimization.</li><li>• Refined evaluation pipelines across foundation models with shot/prompt/inference sensitivity analysis.</li></ul> | <b>Sunnyvale, CA</b><br>2025.7 - present  |
| <b>Amazon AGI Foundations</b><br>Applied Scientist Intern <ul style="list-style-type: none"><li>• Applied parameter efficient LLM training with Quantized LoRA for scalable model adaptation.</li><li>• Devised a model merging framework by distilling activation to accelerate multi-task LLM learning schedule.</li><li>• Implemented adaptive online learning with sample re-weighting for continuous LLM improvement.</li></ul>                                                                                                                                                                                       | <b>Seattle, WA</b><br>2024.7 - 2024.10    |
| <b>University of Massachusetts Amherst</b><br>Research Assistant <ul style="list-style-type: none"><li>• Developed <b>the largest multilingual news similarity dataset</b>, using active learning for retrieval and ranking.</li><li>• Built a SOTA transformer to infer global news graph with trillions of edges for event detection, recommendation.</li><li>• Designed an efficient cluster-aware LLM judge framework for online event extractions from temporal graph.</li></ul>                                                                                                                                      | <b>Amherst, MA</b><br>2019.9 - 2025.6     |
| <b>Z.AI</b><br>Research Scientist (Directly reported to Jie Tang) <ul style="list-style-type: none"><li>• Modeled the dynamics of opinions and personalities with probabilistic graph time series.</li><li>• Devised an extended EM algorithm for learning the graph time series and get competitive performance.</li><li>• Searched and rank scholars with heterogeneous random forest (deployed on prestigious AI platform Aminer).</li><li>• Clustered and visualize temporal scientific topics with refined genetic algorithm.</li></ul>                                                                               | <b>Beijing, China</b><br>2017.6 - 2018.12 |

## PROJECT

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| <b>Nova Foundation Model Training Optimization</b> <ul style="list-style-type: none"><li>• Optimized pre&amp;mid-training recipes for Amazon’s Nova base models, supporting official production release.</li><li>• Designed efficient TB-scale data preparation pipelines to support high-throughput generation.</li><li>• Analyzed training dynamics and failure modes through ablations of hyperparameters and data mixtures.</li></ul> | <b>(Artificial Analysis Intelligence Index: 62)</b> |
| <b>Reasoning Path Diversification in Mid-Trainin</b> <ul style="list-style-type: none"><li>• Rewrote noisy webcrawl data with LLM judge to summarize reasoning information.</li><li>• Designed contrastive knowledge distillation strategies to generate diversified reasoning data.</li><li>• Quantified data reasoning depth with Essential-web classifier to optimize mid-training recipe.</li></ul>                                   | <b>(+16% coding performance)</b>                    |
| <b>Merge LLM with Online Activation Alignment</b> <ul style="list-style-type: none"><li>• Parameter-efficient fine-tuning with data mixing and target evaluation for domain expert models.</li><li>• Developed an activation alignment framework to merge LLMs across diverse tasks (math, coding, medical).</li><li>• Adapted activation weight for LLM online learning during parameter-efficient training and sampling.</li></ul>      | <b>(+10% math performance)</b>                      |
| <b>Cross-Model Evaluation and Benchmarking Framework</b> <ul style="list-style-type: none"><li>• Standardized evaluations across frontier foundation models with shot/prompt robustness check.</li><li>• Enabled rigorous cross-model comparison through unified metrics and analysis workflows.</li></ul>                                                                                                                                | <b>(Aligned up to 17% evaluation gap)</b>           |

- Generated evaluation insights to inform model selection and training roadmap decisions.

### **Efficient Cluster-wise LLM Judge for Event Identifications from Poll** (10x data efficiency)

- Designed a cluster-aware active learning framework to detect temporal events from X poll texts.
- Modeled the evolution of social discourse by graph-based topic tracking and statistical analysis.

### **Global Multilingual News Graph Learning with Transformers**

- Devised a global news retrieval system, generating trillion-scale relations with dense embeddings and indexing.
- Built rank models for similar multilingual news using rules-based and active learning classifiers.

## **SELECTED PUBLICATION**

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### *Amazon Nova 2: Multimodal reasoning and generation models*

Amazon AGI Foundations. Technique Report. 2025.

### *Adaptive Activation Alignment for Model Merging*

Xi Chen, Akshaya Shanbhogue, Yue Zhang. To appear.

### *Identifying and Investigating Global News Coverage of Critical Events Such as Disasters and Terrorist Attacks*

Xi Chen\*, Erica Cai\*, Reagan Keeney, Ethan Zuckerman, Brendan O’connor, Przemyslaw Grabowicz. International AAAI Conference on Web and Social Media (ICWSM), 2025.

### *Efficient Cluster-aware Large Language Model Learning for Event Identification from Social Polls*

Xi Chen, Mayank Bumb, Vishal Kalakonnar, Przemyslaw Grabowicz. To appear, 2025.

### *Global News Synchrony and Diversity During the Start of the COVID-19 Pandemic*

Xi Chen, Scott Hale, David Jurgens, Mattia Samory, Ethan Zuckerman, Przemyslaw Grabowicz. International World Wide Web Conference (The Webconf), 2024.

### *A Multilingual Similarity Dataset for News Article Frame*

Xi Chen, Scott Hale, David Jurgens, Mattia Samory, Przemyslaw Grabowicz. International AAAI Conference on Web and Social Media (ICWSM), 2023.

### *Multilingual News Article Similarity*

Xi Chen, Ali Zeynali, Chico Camargo, Fabian Flock, Devin Gaffney, Przemyslaw Grabowicz, Scott Hale, David Jurgens, Mattia Samory. International Workshop on Semantic Evaluation @ NAACL, 2022.

## **SKILL**

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Pre&mid-training: training dynamics, scaling law, parameter scaling, datamix optimization, synthetic data.

Post-training: Multi-teacher on-policy knowledge distillation, Parameter-efficient fine-tuning, Model merging.

Search & Retrieval: Retrieval-augmented generation(RAG), similarity search, retrieval models, Semantic indexing.

Recommendation Systems: Graph modeling, Active learning for ranking, Multilingual content recommendations.

Machine Learning & Optimization: Event detection, Time-series analysis, Probabilistic modeling.

## **SERVICE**

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### **Invited Talk:**

- Understanding Global Events with LLMs, Pittsburgh, 2026. Host: Ruslan Salakhutdinov & Yaser Sheikh (CMU)
- Multilingual News Article Similarity, Chicago, 2022. Host: Chenghao Tan (UChicago)

### **Program Member:**

- NAACL(2023)
- ICWSM(2023, 2024, 2025, 2026)
- SEMEVAL (2022, 2023, 2024)
- IC2S2 (2022, 2023, 2024, 2025)

## **HONOR**

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NSF Student Travel Award, 2024

UMass CICS best portfolio finalist, 2023.

Nomination for Chinese Exceptional Student (1/6500 ISchool undergrads and grads), 2018.

Ranked global Top50 of Autochess players (50/millions, a world-class strategy game for playing DOTA on chesstable).